

ORIGINAL ARTICLE

Antibacterial metabolites from an unexplored strain of marine fungi *Emericellopsis minima* and determination of the probable mode of action against *Staphylococcus aureus* and methicillin-resistant *S. aureus*

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Abstract

Increasing prevalence of drug resistance has led researchers to focus on discovering new antibacterial agents derived from the marine biome. Although ample studies have investigated marine fungi for their bioactive metabolites with hopeful prospects in drug discovery. The present study was aimed to isolate/ identify potential antimethicillin-resistant *Staphylococcus aureus* compounds producing marine fungal strain from the Indian marine environment. The effective anti-MRSA compound was produced by a marine fungal strain designated as D6. The D6 strain exhibited 99% similarity to *Emericellopsis minima* based on 18S rRNA gene analysis. The culture conditions of *E. minima* D6 were optimized using nutritional and environmental parameters for enhanced anti-MRSA compound production. The agar well diffusion assay was used to determine the inhibition zone diameter of the crude extract against *S. aureus* and methicillin-resistant *S. aureus*, whereas the broth microdilution method was used to determine their minimum inhibitory concentration (MIC) active fraction. MIC values of the ethyl acetate fraction ranged from 0.8 to 1 mg/mL. SEM analysis revealed that the ethyl acetate fraction induces deep craters in methicillin-resistant *S. aureus*. Further, GC-MS analysis confirmed the occurrence of a total of 15 major compounds in active ethyl acetate fraction. Some of the major antibacterial compounds included cyclopentanol, isothiazole, benzoic acid, pyrrolo[1,2-*a*] pyrazine-1,4-dione, and hexahydro. These findings suggest that the marine fungi of *E. minima* can be a valuable candidate for prospecting antibiotics and an alternative complementary strategy for drug-resistant bacterial infections.

KEYWORDS

anti-methicillin-resistant *Staphylococcus aureus*, *Emericellopsis minima*, marine fungi, TLC bioautography

1 | INTRODUCTION

Over the past few decades, the emergence of multidrug resistant pathogens in humans worldwide become a serious concern.¹ Among them, methicillin-resistant *Staphylococcus aureus* (MRSA) is an important nosocomial pathogen within hospitals and in the community. It has been associated with most of the skin and soft tissue infections.² MRSA was first reported in England.³ Annually more than 11,000 deaths were reported due to MRSA-related diseases, including necrotizing pneumonia, osteomyelitis, and sepsis.⁴ MRSA confers resistance to almost all commercially available antibiotics except vancomycin and teicoplanin⁵ leaving them as the only available therapeutics for infections associated with MRSA. However, due to the emergence of vancomycin-intermediate *S. aureus* and vancomycin-resistant *S. aureus*, the novel and potent anti-MRSA compounds are in urgent need to combat the pathogen.

There has recently been increased interest in natural medicine and consequently an increasing need to identify the biologically active compounds from natural resources.⁶ However, getting already known compounds from terrestrial sources, subsequently decreased their potential. Hence, many investigators have been focusing on marine sources for the isolation of unusual natural products with a unique chemical structure. Over the past decade, marine natural products have played an extremely important role in the process of new drug discovery and development.^{7–9} Moreover, from marine sources, many numbers of anti-MRSA metabolites have been isolated from different types of microbes.¹⁰ Marine fungi represent relatively untapped bioresources for novel active components, providing several new metabolites in past years.^{11–13} During this period, approximately 1000 new natural products have been isolated from marine fungi, including terpenoids, alkaloids, shikimate-derived metabolites, lipids, and most importantly polyketides and nonribosomal peptides.^{14,15} Some anti-MRSA compounds were isolated from marine fungi including dehydroxychlorofusarielin B, a polyoxygenated decalin derivative from marine-derived fungus *Aspergillus* sp. with a minimum inhibitory concentration (MIC) value of 62.5 µg/mL against both *S. aureus*, and multidrug-resistant *S. aureus*.¹⁶ The marine-derived fungus *Emericella* sp. was shown to produce mericellamides A and B, which are active against MRSA at MIC values of 3.8 and 6.0 µM, respectively.¹⁷ Likewise, a new antibacterial dioxopiperazine, dehydroxybisdethiobis (methylthio) gliotoxin was isolated from the broth of a marine-derived fungus of the genus *Pseudallescheria* with potent antibacterial activity against the MRSA with MIC values of 31.2 µg/mL.¹⁸ *Emericellopsis* sp. are ubiquitous in the marine

environment and can exist under extreme conditions such as temperature, pressure, salinity, and pH.¹⁹ Only a few reports are available on the isolation of anti-MRSA compounds from *Emericellopsis* sp.^{20–22} and it is yet to be thoroughly investigated for the discovery of novel compounds. Moreover, isolation of anti-MRSA compounds from Indian marine environments has not been reported so far. With this understanding, in the present study, we aimed to identify anti-MRSA compounds producing marine fungi from the Indian marine environment.

2 | METHOD AND MATERIALS

2.1 | Bacterial strain and culture conditions

Both the organisms *S. aureus* (ATCC 11632), which was purchased from the American Type Culture Collection (Manassas, VA, USA), and MRSA, which was obtained from the Regional Institute of Medical Sciences, Imphal, India, were grown at the same culture conditions mentioned by Agrawal et al.²³ with all standard safety precautions.

2.2 | Sample collection and isolation of fungi

Algae and sea sediment samples were collected from a small pit of seawater near the sea of the Dwarkanath Temple (22.23' 77" N × 68.96' 73" E) in Dwarka, Gujarat. All the samples were collected from low tidal areas and stored at 4°C. Isolation of fungi was performed as described by Agrawal et al.²³ Briefly, sample material was inoculated on the isolation media plates amended with chloramphenicol (250 ppm) and incubated at 30°C for 3–4 days to a few weeks until growth was initiated. Distinct fungal colonies on the media plates were then picked and maintained on new potato dextrose agar media plates with artificial seawater.

2.3 | Cultivation of fungal strains for production of active metabolites

A total of 17 morphologically different fungi was isolated from five samples that were collected from Dwarka using PDA media with natural aged seawater. All 17 fungal strains were subjected to liquid fermentation in potato dextrose broth (PDB; Hi-Media, Mumbai, India) as described by Agrawal et al.,²⁴ and the dry crude extracts (mixture



of methanol extract of fermentation broth and fungal mycelia) were obtained and redissolved in methanol to prepare stock solutions of 1 mg/mL and were kept for biological testing.²⁵

2.4 | Screening and identification of potent isolate against MRSA

The agar well-diffusion method described by Perez et al.,²⁶ and Agrawal et al.²⁷ was used to evaluate the antibacterial activity. Briefly, wells of 6 mm diameter were made and filled with 10 μ L of crude extracts (1 mg/mL). The plates were first incubated at 4°C for 15 min and then at 37°C overnight. The diameters of zones of inhibition produced by the test and control antibiotics; streptomycin, ampicillin, and chloramphenicol (1 mg/mL; Sigma-Aldrich, St. Louis, MO, USA) were recorded. For control, pure solvent (methanol) was used. Among these, the D6 isolate showed efficient anti-*S. aureus* and MRSA activity. The morphological and microscopic characteristics of D6 were determined by using compound and scanning electron microscopy (SEM, model EVMA10, Zeiss, Germany) under the supervision of Dr. Sunil Kumar Deshmukh (Fungal taxonomist, India) using various fungal monographs by Beliakova, Benson et al., Backuset al., and Durrel^{28–31} and online database (www.mycobank.org). Subsequently, the 18S rRNA gene sequence of the D6 isolate was submitted to NCBI GenBank under the accession number MF405202.1.

Fungal genomic DNA was extracted from the 7-day-old fungal culture grown in PDB by using minor modification as described by Hermosa et al.³² and Kamala et al.³³ The ITS region of the fungi was amplified using the primer pairs ITS1/ITS4 (5'-TCC GTA GGT GAA CCT GCG G-3'/ 5'-TCC TCC GCT TAT TGA TAT GC-3') and ITS1F/ITS4B (5'-CTTGGTCATTTAGAGGAAGTAA-3'/ 5'-CAGGAGACTTGACACGGTCCAG-3') as described by White et al.,³⁴ in an automated thermocycler (Bio-rad-C1000 Thermal Cycler). The PCR reaction mix of 25 μ L reaction volumes consist of 2.5 μ L of 10 $\times$ buffer, 1.5 μ L of MgCl₂ (25 mM), 0.5 μ L of dNTP (10 mM), 1 μ L of ITS1 primer (5 pm), 1 μ L of ITS4 primer (5 pm), 0.2 μ L of Taq polymerase (2.5 U), 3 μ L of DNA sample (5 μ g/ mL), and 15.3 μ L sterile Milli Q water. The PCR reaction was carried out in a C1000 Touch™ thermal cycler (Bio-Rad, USA) with conditions as follows: denaturation for 5 min at 94°C, 33 elongation cycles (50 s at 95°C, 50 s at 56°C, 1 min at 72°C) final extension for 10 min at 72°C and last at 4°C. Negative controls were used to confirm the absence of contamination. The final products were analyzed in 1% agarose (Sigma-Aldrich) and visualized under UV light using a gel imaging system (BioRad, Chemi Doc, MP). The sequences of the PCR products were analyzed by Finch TV 1.4.0.³⁵

The sequences were then compared with the NCBI GenBank database by the BLASTN program. Phylogenetic relationships were estimated using MEGA 6.³⁶

2.5 | Determination of minimum inhibitory concentration

The MIC of the active fraction was determined by a microdilution assay in 96-well microtiter plates according to the dilution assay according to the EUCAST (<http://www.eucast.org>).³⁷ Streptomycin was used as positive controls. The MIC values were determined at the zero-optical density concentration. All experiments were performed in triplicate.

2.6 | Determination of effects on bacterial cell

Antibacterial activity was further investigated according to Agrawal et al.²³ for their possible effects on targeted cells by scanning electron microscopic analysis. The samples had been scanned under SEM (version EVMA10, Zeiss, Germany) at the TERI-Deakin Nano Biotechnology Centre.

2.7 | Detection of antibacterial compounds by thin-layer chromatography with direct bioautography assay

Thin-layer chromatography (TLC) was performed on aluminum-backed TLC plates (Merck, silica gel 60 F254). A TLC plate loaded with 5 μ L of crude extract (5 mg/mL in methanol) was run in a chloroform: methanol (8:2) solvent system for separation of active principles. TLC plates were further observed under 365 and 254 nm ultraviolet light. The color of resolved bands and R_f values were recorded and marked.³⁸

A contact bioautography against *S. aureus* and MRSA was used as described by Khurram et al.³⁹ The regions of inhibition had been marked and checked with reference TLC, which spot was active. For clean commentary and images of inhibition spots dishes have been stained with methylthiazolyldiphenyl-tetrazolium bromide (MTT) -5 mg/mL-in water.

2.8 | Identification of compound present in the extract

Gas chromatography-mass spectrometry (GC-MS) analysis was done according to previously developed protocol

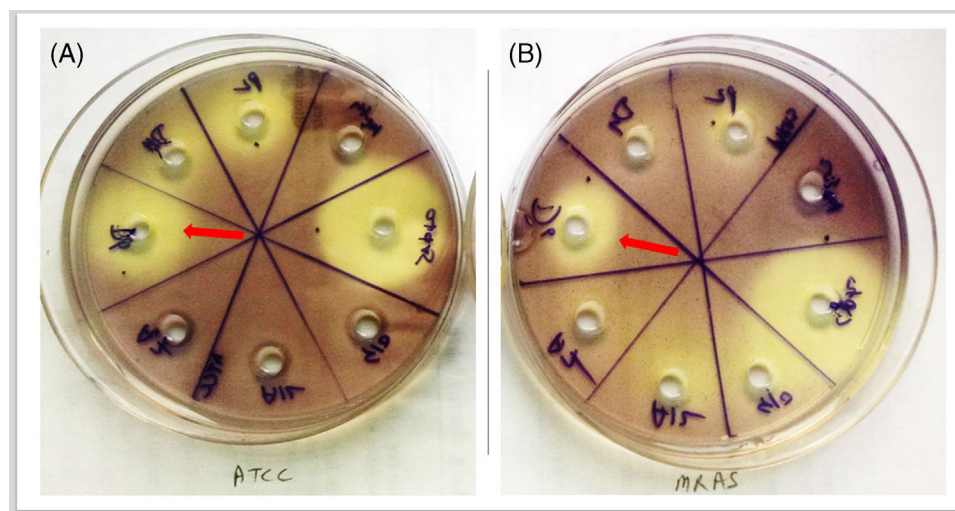


FIGURE 1 Zone of inhibition of *E. minima* (D6) fungal crude extract (red arrow) as determined by the agar well diffusion method against selected bacterial strains *S. aureus* (A) and MRSA (B) along with antibiotic and other fungal crude extracts (clockwise direction after red arrow, D4, D5, methanol, streptomycin, A10, A15, and A7)

by Panigrahi et al.⁴⁰ with the Agilent Technologies 6980N (GC) equipped with a 5975 C mass selective detector (MS), HP-5MS (5% phenyl methyl siloxane) capillary column of dimensions 30.0 m $\times$ 250 μ m $\times$ 0.25 μ m and helium as carrier gas at 1 mL/min. The column temperature was programmed from 50 to 280°C at a rate of 10°C per minute and was kept isothermal for 5 min. The injection port temperature and split ratio were set at 280°C and 1:50, respectively. The MS was operating at 70 eV. The constituents were identified by comparison of their mass spectral data with those from the W9N08 Library.

3 | Results

3.1 | Identification of D6 isolate with anti-MRSA activity

Among the 17 isolates from marine samples collected from Dwarka, the strain D6 showed potent anti-*S. aureus* and MRSA activity with a zone of inhibition of 21 ± 1 mm and 18 ± 1 mm, respectively, using the agar well diffusion assay (Figure 1A,B). The determination of the MIC confirmed that 90–99% inhibition with ethyl acetate fraction at MIC values from 0.8 to 1 mg/mL. The antibacterial activity of the ethyl acetate fraction was markedly lower than that of streptomycin. The D6 was identified by its morphological and sequencing of the rDNA ITS region (ITS1-5.8S-ITS4) as *Emericellopsis minima* (MF405202.1) in BLAST search analysis (Figure 2A–C). We observed ascospores ellipsoid, pale brown, with uneven surfaces, $(4.5\text{--}5.5) \times (2.5\text{--}3.0)$ μ m, surrounded by three, but frequently five longitudinal, subhyaline, smooth-edged alar appendages,

width up to 1.0 μ m. Asexual morph acremonium-like. Conidiation abundant, mostly plectonematogenous, partially hematogenous. Conidiophores are mostly simple orthotropic. Conidiogenous cells are 20–35 μ m long, tapering from 1.5 to 1.8 μ m at the base to 0.7–0.8 μ m at the apex, sometimes lateral branches are formed. Conidia is narrowly ellipsoid, smooth-surfaced, $(3.5\text{--}6.0) \times (1.8\text{--}2.2)$ μ m, about the same length as ascospores but narrower, hyaline, adhering in slimy heads. Chlamydospores are absent.

3.2 | Probable mode of action

The SEM examination of MRSA cells treated and untreated with *E. minima* ethyl acetate fraction is shown in Figure 3A–C. SEM micrographs (Fig. 3a) disclosed that the surface of the untreated MRSA cell was smooth and showed the typical cocci shape. After treatment with ethyl acetate fraction (2 mg/mL) for 18 h resulted in cells showed considerable morphological alterations, including blebs formation on bacterial surfaces, cell wall shrinkage, deformation, and broken cells (Figures 3B,C).

3.3 | Recognition of antibacterial compound in the extract by bioautography

To reveal the antimicrobial constituents by bioautography, 5 mg/mL of D6 crude extract was spotted on TLC. The anti-MRSA bioautography of this TLC clearly showed the presence of more than one number of antibacterial components in the crude extract (Figure 4A). The crude extract was further fractionated with organic solvents

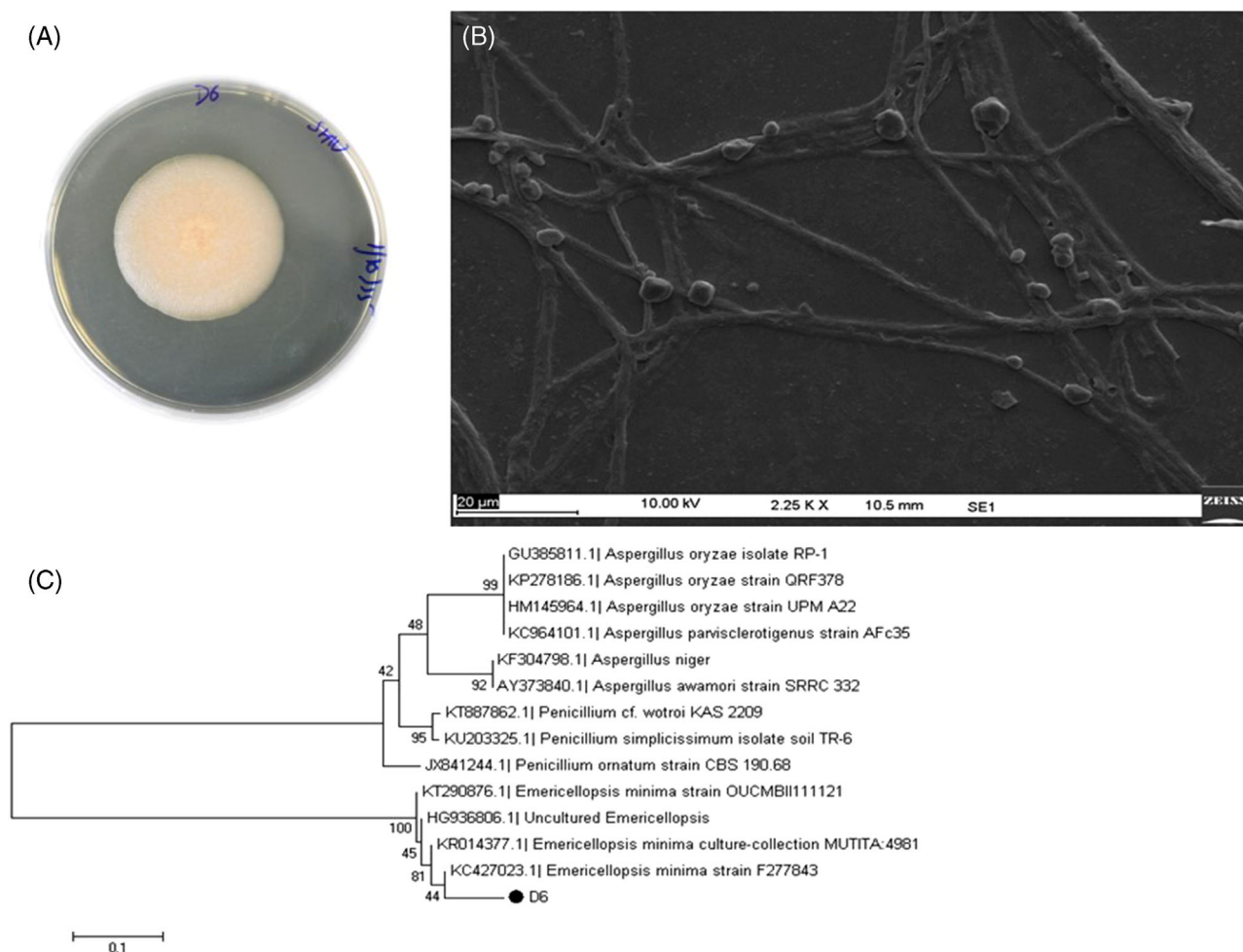


FIGURE 2 Morphological and molecular characteristics of *E. minima* (A) Grown for 10 days on potato dextrose agar, (B) hyphal bundles with acromonium-like conidiation and conidiogenous cells emerging from single hypha (SEM), and (C) phylogenetic tree based on ITS rRNA gene sequence

hexane, chloroform, and ethyl acetate with the help of a separating funnel. We found that the ethyl acetate fraction significantly inhibited the growth of MRSA.

3.4 | GC-MS analysis of *E. minima* strain

GC-MS profiling of the ethyl acetate fraction revealed the occurrence of a total of 25 peaks of which about 15 chemical compounds were identified. These compounds belong to different chemical classes, and most of them are reported to exhibit important biological activities. Many antibacterial compounds like a cyclic alcohol cyclopentanol, an azole derivative Isothiazole, pyrrole derivative pyrrolo[1,2-*a*] pyrazine-1,4-dionehexahydro, and benzoic acid were found to present in ethyl acetate fraction. Further, the presence of cyanoacetamide a fungicide, a vasoconstrictor alkaloid dihydroergotamine and plant growth hormone gibberellin A3 was also observed.

Figure 4B shows the chromatogram of ethyl acetate fraction of *E. minima* crude extract. The identified compounds with their peak number, retention time (RT), and peak area (%) are presented in Table 1.

4 | DISCUSSION

The emergence of multiple drug resistance in human pathogenic microorganisms has become a worldwide health problem. Among them, the rapidly increasing proportion of MRSA is a very serious concern. The thick cell wall of *S. aureus* confers resistance towards all β -lactam antibiotics, including methicillin, oxacillin, penicillin, and amoxicillin.⁴¹ In 2008, about 29% of isolates were of methicillin-resistant *S. aureus* (MRSA), and by 2014, this had risen to 47%. It is also estimated that antibiotic-resistant bacteria are projected to kill 10 million people by 2050.⁴² Antimicrobial resistance has added momentum to

TABLE 1 Chemical composition of ethyl acetate fraction of *E. minima* extract

Number	RT	Name	Area (%)	Class of compound	Molecular formula	Mol. weight	Activity	References
1	3.33	Cyclopentanol	1.45	Cyclic alcohol	C ₅ H ₁₀ O	86	Antibacterial	47
2	3.88	Isothiazole	2.16		C ₃ H ₃ NS	85	Antifungal Antibacterial	53,54
3	4.05	Butanoic acid	2.45	–	C ₄ H ₈ O ₂	88	–	–
4	4.28	Undecane	1.70	–	C ₁₁ H ₂₄	195	–	–
5	4.87	Oxalic acid	1.84	–	C ₂ H ₂ O ₄	90	–	–
6	20.6	3,5-di-tert-Butyl-4-hydroxyanisole	19.78	–	C ₁₁ H ₁₆ O ₂	180	–	55
7	20.8	Benzoic acid	1.67		C ₇ H ₆ O ₂	122	Antimicrobial	[56]
8	21.8	4-Carboxyb enzeneethiol	9.43	–	C ₇ H ₇ BO ₄	165	–	–
9	21.9	Pyrrolo[1,2-a]pyrazine-1,4-dione, hexahydro	13.39	–	C ₇ H ₁₀ N ₂ O ₂	154	Anti -MRAS Antioxidant	46,57
10	22.0	4-Methoxythioanisole	1.13	–	C ₈ H ₁₀ OS	154	–	–
11	23.1	Cyanoacetamide	5.60	Acetic amide	C ₃ H ₄ N ₂ O	351	Fungicide	58
12	24.2	Benzene propane nitrile	19.8	Nitrile	C ₉ H ₉ N	131	–	–
13	26.1	Dihydroergotamine	2.57	Alkaloid	C ₃₃ H ₃₅ N ₅ O ₅	581	vasoconstrictor	59
14	27.9	Gibberellin A3	1.20	Hormone	C ₁₉ H ₂₂ O ₆	346	Plant growth promoter	60
15	28.0	Sillicic acid	2.53	Weak acid	C ₁₀ H ₂₈ O ₄ Si ₃	296	–	–

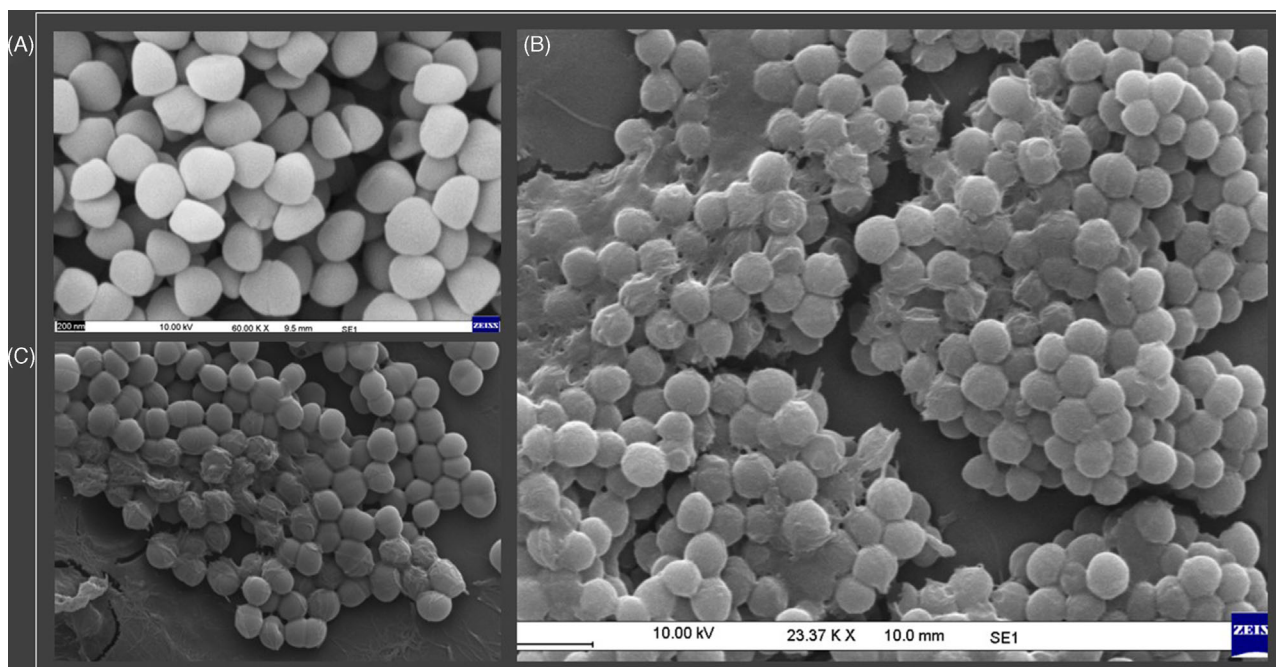


FIGURE 3 SEM analysis of MRSA after treating with 2 mg/mL of the *E. minima* ethyl acetate fraction for 18 h. (A) Control and (B, C) indicates deformation in cell morphology and finally rupturing in cell membrane leading to leakage of the DNA, RNA, and proteins through the membrane

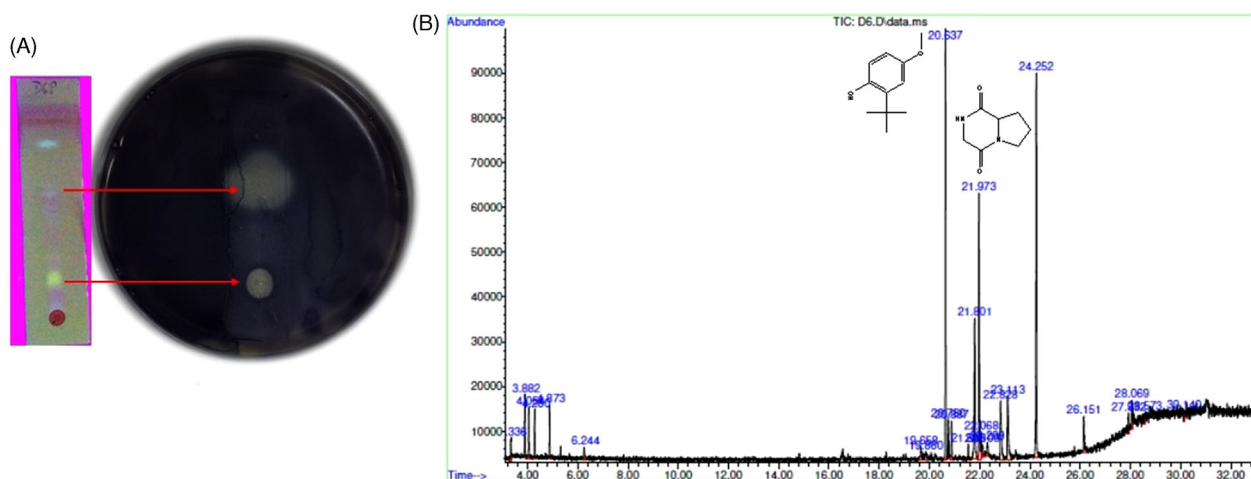


FIGURE 4 (A) TLC bioautography based recognition of active molecules in crude extract (B) The GC-MS-based chemical profiling of ethyl acetate fraction of *E. minima* crude extract

exploration for new antimicrobial substances from natural sources. As natural products have a wide range of structural diversity and history of discovery as new and novel antibiotics and chemotherapeutic agents. Due to chemical complexity, many times crude extracts show more effectiveness than chemically synthesized pure compounds. Their complexity makes them to interact with multiple molecular targets thus conveying accrued difficulty for tar-

get microorganisms to develop resistance due to multiple response sites.^{43,44} In between 1940 and 1962, more than 20 new antibiotics were discovered and marketed; thereafter, the discovery and marketing of new compounds slackened. Rigorous efforts in the past decades have led to the discovery of promising antimicrobial extracts and compounds from marine microorganisms. In particular, marine bacteria and fungi have been recently the subject

of several antimicrobial screening studies worldwide.^{14,45} Here, we present data indicating the antibacterial efficacy and antibiotic potentiating activity of crude extracts and fractions obtained from marine fungi from the Indian marine area. In this context, the sea sediment and algae samples were collected from the different locations of the Dwaraka coastline for the isolation of fungi. In our study, a total of 17 marine fungi morphologically different strains were isolated and were preliminarily screened against *S. aureus* and MRSA. Among them, ethyl acetate extract of 15-day-old fermented broth of strains D6 strains showed the most significant activity against both pathogens. GC-MS analysis revealed the presence of known anti-MRSA compound pyrrolo[1,2-*a*] pyrazine-1,4-dione, hexahydro, which showed a potent inhibitory effect on multidrug-resistant *S. aureus* with a MIC of 15 ± 0.172 mg/L and minimum bactericidal concentration of 20 ± 0.072 mg/L. The compound was nonhemolytic and also showed high antioxidant activity.⁴⁶ Cyclopentanol, which was previously reported as a moderate inhibitor of *S. aureus* with the MIC of 50 μ g/mL concentration.⁴⁷ Thiazole, which showed anti-MRSA activity at the MIC value of 0.25–8 mg/L⁴⁸ and benzoic acid, is a well-known antibacterial compound with a MIC value of >500 μ mol/L.⁴⁹

Previous studies on *E. minima* have already established it as the reservoir of antimicrobial compounds.^{12,75,76} The results of our study are in agreement with the findings of Inostroza et al.⁵⁰ who have reported antibacterial of *E. minima* against multidrug-resistant bacteria. They isolated emerimicin IV,⁵¹ a unique fungal peptaibol that exhibited bacteriostatic activity against clinical isolates of methicillin-resistant *S. aureus* and vancomycin-resistant *Enterococcus faecalis* with MIC values ranging between 100 and 12.5 μ g/mL. Likewise, Rogozhin et al.⁵² isolated a novel lipopeptaibol Emericellipsin A from an alkalophilic *E. alkalina*, which showed antibacterial activity against *S. aureus* FDA 209 P with a MIC of 4 μ g/mL. Further, bioassay-guided fractionation of the active fraction along with LC-MS analyses will also be needed to isolate and identify the other active constituents (undetected by GC-MS) contributing to the antibacterial property of extract and elucidate the mode of action of pure compounds.

5 | CONCLUSIONS

In this study, we have successfully isolated and characterized a marine derive fungal strain *E. minimum* and produce antibiotic compounds that acted on MRSA. We also proposed that the primary mechanism of the extract action is associated with the disruption of the bacterial cellular membrane. Advocating the possibility of finding potent antibacterial agents, particularly from its ethyl acetate frac-

tion. The spectroscopic analysis revealed the anti-MRSA bioactive metabolite as a pyrrolo[1,2-*a*] pyrazine-1,4-dione, hexahydro. In this perspective, it was concluded that this marine fungus can be used as a sustainable bioresource for the production of pyrrolo[1,2-*a*] pyrazine-1,4-dione, hexahydro, which could be developed as an effective antibiotic against MRSA.

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AUTHOR CONTRIBUTIONS

Conceived and designed the experiments: SA; Performed the experiments: SA; analyzed the data: SA, LD, SD; contributed reagents/materials/analysis tools: SA; wrote the paper: SA; editing: LD, SD.

CONFLICT OF INTEREST

The authors have no conflict of interests to declare.

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